

NUMERICAL METHODS FOR PDES

Lecture 1

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LECTURES

All lectures in room A-133 and Teams

- Tue 11-13
- Wed 16-18

Course material: <https://github.com/andreacangiani/NMPDE2026>

ASSESSMENT:

By oral exam. Just before your oral, you will be asked to attempt a written theoretical exercise.

Admission to exam if computer classes completed.

IDEA OF THE COURSE:

- Physics-based main problems and methods
- Fundamental ideas/methods and the interlink between approaches.
- All main classes of linear PDEs (elliptic, parabolic, hyperbolic).
- Choice of appropriate methods depend on the problem
- Mainly theoretical, with python classes on basic implementation

BIBLIOGRAPHY:

- Lecture notes
- Morton & Mayers *Numerical Solution of Partial Differential Equations*. Cambridge, 1994. (MM)
- Larsson & Thomee *Partial Differential Equations with Numerical Methods*. Springer, 2009. (LT)
- Quarteroni *Numerical Models for Differential Problems*. Springer, 2017. (Q)

1.1 well-posedness + reliable discretization

(Q 1.2) $X \rightarrow Y$

Let $P(u, g) = 0$ a PDE problem
 $\downarrow \quad \downarrow$
 solution data

resolvent

$\downarrow$

- $\exists!$ solution $u \rightarrow u = R(g)$
- u depends continuously on data g

$\hookrightarrow$ Denote by δg admissible perturb. of the data

δu change in the solution

$$\Rightarrow P(u + \delta u, g + \delta g) = 0$$

$$\Rightarrow \forall \eta > 0, \exists K(\eta, g) < \infty : \|\delta g\| < \eta$$

$$\Rightarrow \|\delta u\|_X \leq K(\eta, g) \|\delta g\|_Y$$

$$K = \sup_{\substack{Sg \in B \\ \text{(adm. pert.)}}} \frac{\|Su\|_X / \|u\|_X}{\|Sg\|_Y / \|g\|_Y}$$

condition
number of
the PDE
problem

Numerical method :

$$P_h(u_h^{EXh}, g_h^E) = 0$$

g_h = approximation of the data

h = resolution parameter

family
(h)

stability : The family $P_h(u_h, g_h) = 0$ is stable if small perturbations in g_h result in uniformly small perturbations in u_h .

that is $\forall \varepsilon > 0 \exists \delta_\varepsilon > 0$: if $\|Sg_h\| < \delta$

then $\|Su_h\| < \varepsilon$
($\forall h$)

example (exercise) : suppose

$$A_h u_h - f_h = 0$$

then stability $\Leftrightarrow \|A_h^{-1}\| \leq C$
 C indep. of h .

Consistency: family of num. methods is
consistent if (assume that P_h can be
tested with u)
$$\tau_h := P_h(u, g_h)$$
 TRUNCATION
ERROR

$$\downarrow \text{ as } h \rightarrow 0$$
$$0$$

and strongly (fully) consistent

$$\tau_h \equiv 0$$

CONVERGENT: family of numerical methods
is convergent if $\lim_{h \rightarrow 0} \|u - u_h\|_X = 0$

Example: $A_h u_h = f_h$ ($\swarrow$ $I_h u = \text{interpolation}$)
assume as above that $A_h u = f_h$ makes sense, then $\tau_h = A_h u - f_h$

$$\Rightarrow A_h u = f_h + \tau_h$$

If $\|A_h^{-1}\| \leq C$ uniformly (stability)

Then $\|u - u_h\| \leq C \|\tau_h\|$

$\downarrow$ (consistency)
0

convergence

Theorem (Lax-Richtmyer equivalence):
for linear PDE problems
consistency + stability $\Rightarrow$ convergence

Consider 1D Poisson problem

$$\begin{cases} -u'' = f & \text{in } (0, 1) = \Omega \\ u|_{\partial\Omega} = g \end{cases}$$

Def : $u : \Omega \rightarrow \mathbb{R}$ is harmonic if $\subset \mathbb{R}^d$

$$\{u \in C^2(\Omega)$$

Ω open

Laplace $\rightarrow$

$$\Delta u = 0 \quad \text{in } \Omega$$

Eq.

$\hookrightarrow$ Laplacian in $\mathbb{R}^d$ $\Delta u = u_{xx} + u_{yy} + \dots$

Theorem (mean value formula)

$u \in C^2(\Omega)$ is harmonic $\Leftrightarrow$

$$u(x) = \int_{\partial B_r(x)} u \, ds$$

$$\oint_{\omega} u = \frac{1}{|\omega|} \int_{\omega} u$$

for each ball : $\overline{B_r(x)} \subset \Omega$.

We also have

$$u(x) = \int_{B_r(x)} u \, dS$$

Theorem (Strong Maximum Principle)

Let $u \in C^2(\Omega) \cap C(\bar{\Omega})$ harmonic in Ω
then

$$(i) \quad \max_{\bar{\Omega}} u = \max_{\partial\Omega} u$$

$$\min_{\bar{\Omega}} u = \min_{\partial\Omega} u$$

(ii) If Ω is connected and

$$\exists x_0 \in \Omega : u(x_0) = \max_{x \in \bar{\Omega}} u$$

then $u = \text{const}$ in Ω

proof: (i) let $x_0 \in \Omega$, $u(x_0) = M = \max_{\bar{\Omega}} u$

then for $B_{x_0}(r) \subset \Omega$ we have

$$M = u(x_0) = \int_{B_{x_0}(r)} u \leq M$$

! mean-value

$$\Rightarrow u \equiv \Pi \quad \text{in } B_{x_0}(r)$$

$$\text{Define } D := \{x \in \Omega : u(x) = \Pi\}$$

D is — open
— (relatively) closed in Ω

$$\Rightarrow D \equiv \Omega.$$

! Ω connected

Corollary (uniqueness) : Let $g \in \mathcal{C}(\partial\Omega)$
 $f \in \mathcal{C}(\Omega)$. Then there exists at
most one solution $u \in \mathcal{C}^1(\Omega) \cap \mathcal{C}(\bar{\Omega})$

of

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega \end{cases} \quad (*)$$

proof (exercise using S.M.P)

alternative proof by Energy method:

suppose $u, \tilde{u}$ two solutions, set

$$w := u - \tilde{u}$$

then : $\bullet$ w is harmonic

$$\bullet w|_{\partial\Omega} = 0$$

$$0 = - \int_{\Omega} w \Delta w \stackrel{\text{Gauss-Green}}{=} \int_{\Omega} |\nabla w|^2 - \cancel{\int_{\partial\Omega} w^2} \stackrel{0}{=} 0$$

$$\Rightarrow \begin{cases} \nabla w \equiv 0 \\ w|_{\partial\Omega} \equiv 0 \end{cases} \Rightarrow w \equiv 0$$

$\bullet$ existence $\rightarrow$ by construction (Green's functions)

o Well-posedness:

Theorem: Let $u \in \mathcal{C}^1(\Omega) \cap \mathcal{C}(\bar{\Omega})$ solve
(*) , then

$$\|u\|_{L^\infty(\bar{\Omega})} \leq \|g\|_{L^\infty(\partial\Omega)} + C \frac{\|f\|_{L^\infty(\Omega)}}{\Delta u}$$

Corollary $\rightarrow$ sol depends
continuously
on data

Back to 1D case

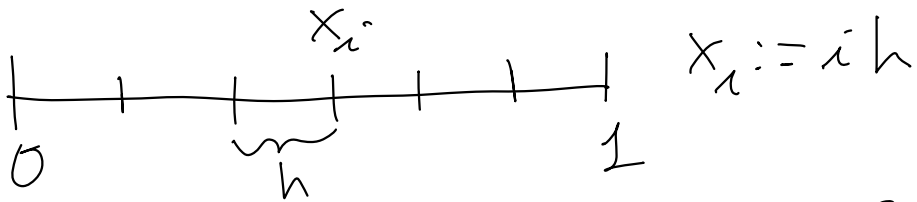
$$\begin{cases} -u'' = f & \Omega = (0, 1) \\ u(0) = 0 = u(1) \end{cases}$$

$$\mathcal{L}u := -u''$$

Finite Difference Method

- Fix a subdivision of Ω

- Fix N $h = \frac{1}{N}$



- Approximate u by $u_h = \left\{ U_i \right\}_{i=0}^N$
where $U_i \approx u(x_i)$

- Approximate u'' by second
central divided difference

$$u''(x) \approx \delta_h^2 u(x) = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

why?

$$\underbrace{\int_h u(x)}_{\text{central divided diff. op.}} := \frac{u(x+h/2) - u(x-h/2)}{h}$$

$$\begin{aligned} & \int_h^2 \\ & \parallel \\ \int_h \left(\int_h u(x) \right) &= \int_h \left(\frac{u(x+h/2) - u(x-h/2)}{h} \right) \\ &= \frac{1}{h} \left(\int_h (u(x+h/2)) - \int_h u(x-h/2) \right) \\ &= \frac{1}{2h} \left(\frac{u(\overset{\text{red } x_i}{x+h}) - u(x)}{h} - \frac{u(x) - u(x-h)}{h} \right) \end{aligned}$$

FDM: Find $u_h = \{U_i\}_{i=0}^M$:

$$\mathcal{L}U_i := \begin{cases} -\int_n^2 U_i = f_i & \forall i=1, \dots, N-1 \\ U_0 = 0 = U_N \end{cases}$$

Numerical Analysis

DMPP

Theorem (discrete maximum principle)
 Let $V = \{V_i\}_{i=0}^N$. If $\mathcal{L}_h V_i = -\int_n^2 V_i \leq 0$
 $\forall i=1, \dots, N-1$
 then $\max_i V_i = \max \{V_0, V_N\}$

Proof: suppose $V_n = \max_{1 \leq i \leq N-1} V_i$, $\begin{matrix} V_n > V_0 \\ \text{or} \\ V_n > V_N \end{matrix}$

$$\text{then } 0 \geq h^2 \int_n^2 V_n = \cancel{h^2} \frac{V_{n+1} - 2V_n + V_{n-1}}{\cancel{h^2}}$$

$$\Rightarrow V_n \leq \frac{V_{n+1} + V_{n-1}}{2} \leq V_n$$

$$\Rightarrow V_n = V_{n+1} = V_{n-1}$$

apply again until reaching the boundary $\Rightarrow V$ constant vector

Lemma (stability): For any

$$V = \{V_i\}_{i=0}^H$$

$$\|V\|_{\ell^\infty} \leq \max\{|V_0|, |V_H|\} + \frac{1}{8} \|\Sigma_h V\|_{\ell^\infty}$$

Proof (tomorrow)

(In particular, for $U=V$ $\Sigma_h U = F$

$$\text{where } F = \left\{ f(x_i) \right\}_{i=1}^{H-1}$$

$\rightarrow$ continuous dependence on data for the FDM.)

Corollary: The discrete FDM problem has at most one solution.

$\Rightarrow$ Hence it is well-posed